

# Introduction to ASP.NET

## What is ASP.NET?

ASP.NET is a web application framework developed by Microsoft.

- Used to create dynamic websites, web applications, and web services.
- Runs on the server side and generates HTML sent to the client browser.
- Common programming languages: C# and VB.NET.

## The .NET Framework

A software development platform for building and running applications.

### Key components:

- **Common Language Runtime (CLR)** – manages memory, execution, and security.
- **.NET Class Library** – reusable classes for data access, file handling, web services, etc.

ASP.NET is part of the .NET ecosystem, focused on web development.

## ASP Classic vs ASP.NET

| Feature           | ASP Classic          | ASP.NET                  |
|-------------------|----------------------|--------------------------|
| Release Year      | 1996                 | 2002                     |
| Programming Model | Script-based         | Compiled                 |
| Languages         | VBScript, JavaScript | C#, VB.NET               |
| Performance       | Slower               | Faster (precompiled)     |
| Code Structure    | HTML and logic mixed | Supports code-behind     |
| Extensibility     | Limited              | Uses full .NET libraries |

## Server Services in ASP.NET

ASP.NET provides built-in services such as:

- State Management (ViewState, Session, Application)
- Security and Authentication
- Caching and Performance Optimization
- Compilation and Error Handling
- Integration with .NET libraries

## Key Features of ASP.NET

- Extensibility
- High performance and scalability
- Code-Behind architecture
- Dynamic pages
- Cross-platform compatibility

## References

- <https://learn.microsoft.com/en-us/aspnet/overview>
- [GeeksforGeeks: Introduction to ASP.NET](#)